

Interview Transcript with Dr. Michael Anderson (New York, NY) - ICU Director

Interviewer: Thank you for joining us, Dr. Anderson. To start, can you introduce yourself and describe your role during the COVID-19 pandemic? Dr. Anderson: Thank you for having me. I am Michael Anderson, an intensivist by training and the ICU Director at a large tertiary academic hospital in New York City. I have been practicing critical care for eighteen years. During the pandemic, my responsibilities extended beyond bedside care to include rapid operational scaling of ICU capacity, triage protocol development, leading multi-disciplinary critical care committees, and coordinating with hospital leadership and regional health authorities on resource allocation and data-driven clinical pathways.

Interviewer: When did your hospital first start seeing COVID-19 patients and what were those initial weeks like? Dr. Anderson: Our hospital saw the earliest sustained surge starting in March 2020. I can still recall the first week when the emergency department began filling with patients with severe hypoxemia. The initial weeks were marked by uncertainty, fear, and improvisation. We had to convert operating rooms and step-down units into negative-pressure areas, repurpose anesthesia machines into ventilators in some cases, and develop makeshift donning and doffing stations. Supply chains for N95s and gowns were strained, and we had to institute conservation strategies. Clinically, we were dealing with an unfamiliar disease with rapidly changing guidance; every evening brought new research, and every morning demanded adaptation.

Interviewer: Clinically, what patterns did you see that were distinct or surprising? Dr. Anderson: One of the most startling patterns early on was "silent hypoxia," where patients presented with dangerously low oxygen saturations but relatively little subjective dyspnea. We also observed a high incidence of thrombotic events - deep vein thromboses, pulmonary emboli, and clotting of dialysis and vascular access lines - which prompted us to add anticoagulation strategies into our care pathways. Beyond the lungs, cardiac injury and renal dysfunction were frequent; some patients developed myocarditis or required renal replacement therapy. The heterogeneity in presentation - from mild disease to sudden multi-organ collapse - made prognostication difficult and forced us to be conservative with resources while concurrently searching for markers that would help predict deterioration.

Interviewer: How did your ventilatory and respiratory support strategies evolve as you learned more? Dr. Anderson: At the start, many centers, including ours, favored early intubation out of concern for aerosolization and fast deterioration. But as colleagues reported experience and we reviewed accumulating data, the practice evolved. We began to deploy high-flow nasal cannulae more liberally in monitored settings, used non-invasive supports such as helmet CPAP for select patients, and pursued awake proning even for those on supplemental oxygen. For ventilated patients, lung-protective strategies (low tidal volumes, conservative driving pressures) became standard, and we became prudent with sedation and paralysis. We also learned that timing and

patient selection mattered: some patients benefited from non-invasive support while others required prompt escalation. The nuance developed over months of shared learning across the system.

Interviewer: Can you describe the operational challenge of expanding ICU capacity and managing workforce during surges? Dr. Anderson: Scaling critical care capacity meant more than beds and ventilators - trained staff were the true constraint. We expanded ICU capacity by repurposing PACU and stepdown units, but then faced a shortage of ICU nurses and respiratory therapists. The solution involved tiered staffing models: experienced ICU nurses supervised teams that included less-experienced staff, we redeployed anesthesiologists and surgical intensivists, and we used tele-ICU to extend expertise across sites. Rapid orientation and simulation training were conducted to upskill staff, but even so, maintaining quality under pressure was a daily achievement. Staff wellbeing was prioritized with rotations to limit continuous exposure, dedicated rest areas, and mental health resources; even with these supports, the moral burden of repeated deaths was profound and had lasting effects.

Interviewer: Triage and allocation became necessary in some settings. How were these decisions made ethically and practically? Dr. Anderson: We established an independent triage committee that included clinicians, ethicists, palliative care specialists, and administrators. The committee used transparent, medical-score-based criteria to evaluate likelihood of benefit, comorbid conditions, and reversibility rather than age alone. The process emphasized fairness and consistency, and we communicated the framework publicly. It was critical to have separation between bedside teams and allocation decision-makers to preserve trust and to reduce moral injury among frontline providers who would otherwise bear impossible choices in the moment.

Interviewer: Can you recount any specific patient stories that shaped policy or practice in your ICU? Dr. Anderson: One case that resonates involved a middle-aged nurse who presented with mild symptoms and then deteriorated rapidly over 36 hours, developing thrombotic complications and multi-organ failure. Her course underscored the need for early thromboprophylaxis adjustments and the importance of monitoring inflammatory markers like D-dimer and ferritin as adjuncts to clinical judgment. Her case, and several similar ones, pushed us to incorporate more aggressive monitoring and to set early thresholds for escalation, and it accelerated our participation in multicenter trials focusing on anticoagulation strategies.

Interviewer: What role did clinical trials and evidence synthesis play on the floor? Dr. Anderson: Clinical trials were pivotal. We tried hard to enroll patients in randomized protocols whenever feasible because the pandemic generated many plausible interventions that required rigorous evaluation. At the institutional level, we ran rapid "watch committees" that reviewed incoming preprints and peer-reviewed studies daily. Protocol changes went through quick feedback cycles with bedside staff so that changes were implementable. Trials that later demonstrated benefit, such as dexamethasone for hypoxic patients, were rapidly adopted and

became part of our standardized order sets.

Interviewer: How did communication with families and visitor policies evolve, given infection control constraints? Dr. Anderson: Restricted visitation presented a heartbreaking trade-off. We implemented structured virtual visitation programs - tablets and secure video platforms - to allow families to see and speak with patients. Palliative care was integrated early for patients with poor prognoses, and we trained staff in compassionate communication over video calls. We also made exceptions for end-of-life visits with strict PPE and limited numbers where infection risks were mitigated, because human connection is crucial for families and staff closure.

Interviewer: Did technology such as tele-ICU and AI play sustainable roles, or were they emergency measures only? Dr. Anderson: Tele-ICU proved sustainable and transformative in extending intensivist expertise to smaller hospitals in our network. It allowed real-time monitoring and consultant support, which reduced unnecessary transfers and improved local care. AI tools for early warning signals were piloted; while promising, many required local calibration and validation before being trusted. In short, telehealth modalities became integral whereas AI remained an assistive tool pending further validation.

Interviewer: How did vaccination change admission patterns and your clinical workload? Dr. Anderson: Vaccination dramatically reduced severe cases among fully vaccinated patients, shifting the ICU case mix toward unvaccinated individuals and those with poor vaccine responses, like certain immunocompromised patients. This shift reduced peak occupancy pressures in later waves and allowed us to reintroduce elective care. However, vaccine hesitancy in some communities continued to drive preventable severe disease, which was frustrating for clinicians who witnessed the difference vaccination could make firsthand.

Interviewer: What about post-discharge care and long-COVID - how has your unit engaged with recovery pathways? Dr. Anderson: We established multidisciplinary post-COVID clinics involving pulmonology, cardiology, neurology, and rehabilitation medicine. Many survivors had prolonged dyspnea, cognitive deficits, and fatigue; early rehabilitation, pulmonary function testing, and coordinated follow-up reduced readmissions and improved quality of life. Designing these clinics required redirecting outpatient resources and ensuring equitable access for lower-income patients who often faced the worst long-term sequelae.

Interviewer: Looking forward, what structural changes would you prioritize to improve preparedness for future pandemics? Dr. Anderson: Invest in workforce development and retention, create flexible surge-capable staffing models with cross-training, ensure regional coordination for critical supplies such as oxygen, establish rapid-response trial networks, and normalize mental health support for clinicians. Moreover, strengthening primary prevention and public health infrastructure to reduce preventable surges will always be more cost-effective than crisis response.

Interviewer: Any final reflections on the human dimension of the pandemic response? Dr.

Anderson: Beyond protocols and equipment, the pandemic revealed the extraordinary courage and compassion of healthcare teams. It also exposed systemic weaknesses that can and must be addressed. Honoring that dedication means committing to long-term improvements in policy, investment, and societal support for healthcare systems.

Interviewer: Thank you for your time and insights. Dr. Anderson: Thank you. My hope is that the lessons learned translate into durable change so that the next crisis finds us better prepared, more resilient, and more compassionate. --- End of Interview ---